ALGEBRA 2 · CONIC SECTIONS

Conic Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Conic Explorer** on Mr. K's Classroom: Algebra 2 → Conic Sections → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. What do you think happens to an ellipse when a = b?

   It becomes a circle.

2 Explore use the tool and record what you see

Identify and describe

Build each conic in the tool.

Equation	Conic	Center or vertex	Key facts
$(x-2)^2+(y+1)^2=25$	**circle**	**(2, –1)**	**radius 5**
$\frac{x^2}{25}+\frac{y^2}{9}=1$	**ellipse**	**(0, 0)**	**horizontal major axis; foci (±4, 0)**
$\frac{x^2}{9}+\frac{y^2}{25}=1$	**ellipse**	**(0, 0)**	**vertical major axis; foci (0, ±4)**
$\frac{x^2}{4}-\frac{y^2}{16}=1$	**hyperbola**	**(0, 0)**	**opens left and right; asymptotes y = ±2x**
$y^2=-8x$	**parabola**	**(0, 0)**	**opens left; focus (–2, 0); directrix x = 2**
$(x-1)^2=4(y+3)$	**parabola**	**(1, –3)**	**opens up; focus (1, –2); directrix y = –4**

3 Write the rule fill in each blank from what you observed

1. A circle with center (h, k) and radius r: $(x-h)^2+(y-k)^2=$ **r^2**.
2. For an ellipse, $c^2=$ **a^2-b^2 (larger minus smaller)**. For a hyperbola, $c^2=$ **a^2+b^2**.
3. In an ellipse, the larger denominator tells the direction of the **major** axis.
4. The asymptotes of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ have slopes **$\pm\frac{b}{a}$**.
5. Every point on a parabola is the same distance from the **focus** and the **directrix**.

4 Check no tool: show your work

1. Center and radius of $x^2+(y-3)^2=49$?

   (0, 3); 7

2. Foci of $\frac{x^2}{100}+\frac{y^2}{36}=1$?

   (±8, 0)

3. Asymptotes of $\frac{y^2}{9} - \frac{x^2}{4} = 1$?

$y = \pm\frac{3}{2}x$

4. Focus of $x^2 = 12y$?

(0, 3)

5 Reflect

How can you tell an ellipse from a hyperbola by looking at the equation?

An ellipse adds the squared terms; a hyperbola subtracts them.